Extraction from Operational Sources to Staging

Each numbered DFT runs separately; the lower lanes explain the data flow pattern.

CONTROL FLOW - THE SAME SEQUENTIAL ORDER AS THE SSIS PACKAGE

SQL_Clear_Staging
Clear all 12 staging tables before extraction

→

12 DATA FLOW TASKS - RUN IN NUMBERED ORDER

01. DFT_Extract_Customer
02. DFT_Extract_Country
03. DFT_Extract_BookLanguage
04. DFT_Extract_OrderStatus
05. DFT_Extract_AddressStatus
06. DFT_Extract_Address
07. DFT_Extract_CustomerAddress
08. DFT_Extract_Book
09. DFT_Extract_CustOrder
10. DFT_Extract_OrderLine
11. DFT_Extract_OrderHistory
12. DFT_Extract_CustomerSurvey

SOURCE SYSTEMS

SSIS DATA FLOW - TWO INDEPENDENT LANES

STAGING TABLES

gravity_books
customer, country
book_language, order_status
address_status, address
customer_address, book
cust_order, order_line
order_history

→

OLE DB Source
Select required columns
CM_Gravity_Source

→

Derived Column
LoadBatchID =
@[User::LoadBatchID]

→

OLE DB Destination
Fast load matching stg table
CM_GravityDW

→

GravityDW.stg
Customer, Country
BookLanguage, OrderStatus
AddressStatus, Address
CustomerAddress, Book
CustOrder, OrderLine
OrderHistory

SURVEY LANE: one independent data flow

Bookstore survey CSV
Raw survey responses
CM_Survey_File

→

Flat File Source
Read survey fields
as raw text

→

Derived Column
LoadBatchID =
@[User::LoadBatchID]

→

OLE DB Destination
Fast load raw survey fields
CM_GravityDW

→

stg.CustomerSurvey
CustomerIDRaw
DiscoverySourceRaw
PurchasePreferenceRaw

Extraction metadata
Shared positive LoadBatchID is supplied manually.
StagingRowID is generated by SQL Server.
ExtractedAt defaults to SYSUTCDATETIME().

Separate SSMS profiling
No source-ID unique or foreign-key constraints in staging.
Profile counts, IDs, missing values and source links separately.
These checks are not extra tasks in the SSIS control flow.
The SQL and CSV lanes do not merge during extraction.

→ Success / execution order

→ Data movement

Figure 1 / Extraction to staging

SQL Preparation and Warehouse Loading

Numbered panels summarise sequential tasks, not SSIS containers. SQL views are SELECT definitions, not extra stored tables.

1 / Reference dimensions

1. SQL_Initialize_Warehouse
2. SQL_Update_Country_And_Status
3. DFT_Load_DimDeliveryCountry
4. DFT_Load_DimOrderStatus

2 / Book and customer dimensions

5. SQL_Update_DimBook
6. DFT_Load_DimBook
7. SQL_Update_DimCustomer
8. DFT_Load_DimCustomer

3 / Validate and load facts

9. SQL_Validate_And_Update_FactOrderLine
10. DFT_Load_FactOrderLine

All tasks run sequentially on success.

INPUTS

SQL PREPARATION AND MAINTENANCE

WAREHOUSE TARGETS

Generated calendar

2023-01-01 to 2026-12-31
Not a staging date table

SQL_Initialize_Warehouse

Generate dates; DateKey = YYYYMMDD.
Ensure technical unknown members exist.

dw.DimDate

Country + OrderStatus

Reference data from staging

Country and status maintenance

Update labels by source ID; insert unseen IDs.
Existing surrogate keys remain stable.

dw.DimDeliveryCountry

dw.DimOrderStatus

Book + BookLanguage

Book and language attributes

etl.vwBookPrepared

Trim / cast text; classify reviewed language codes.
English | Historical English | Multilingual | Non-English | Unknown / needs review
Type 1 updates; insert unseen source IDs.

dw.DimBook

Customer inputs

Customer + CustomerSurvey
CustomerAddress + Address
Country

etl.vwCustomerPrepared

TRY_CONVERT survey IDs; standardise answers; retain nonrespondents.
Aggregate distinct active country IDs before the customer join.
Active = status_id 1, verified using stg.AddressStatus during profiling.

dw.DimCustomer

Order-line inputs

OrderLine + CustOrder
OrderHistory + Address
OrderStatus supports validation

etl.vwFactOrderLinePrepared

Join orders, lines and destinations.
Determine latest recorded status.
Resolve all 5 dimension keys in SQL.
One prepared row per source line.

Validate, update, then insert

etl.uspValidateAndUpdateFactOrderLine
@ApplyUpdates = 1
Update changed existing facts.
DFT_Load_FactOrderLine inserts new IDs.

dw.FactOrderLine

One row per source order line
SourceOrderLineID is unique
OrderLineCount = 1 counts lines

Fact insertion data flow: SRC_Prepared_FactOrderLine -> DST_FactOrderLine

Keys and repeat loads

Generate identity keys for new members.
Retain existing keys; unknown member = 0.
DimDate uses YYYYMMDD.
Customer attributes use Type 1 updates.

Preserved exceptions

3 orders without history use unknown status.
Retain cancelled and returned orders.
Preserve 3 recorded zero-price lines.
Missing survey responses remain explicit.

Validation failures

Check shared batch, grain, links, dates and prices.
Conflicting latest statuses require review.
SQL errors block dependent fact insertion.
Earlier successful tasks may have committed.

Success / execution order

Data movement

Figure 2 / Transformation and loading